

Q.24 What is HMI? What are the properties of it? How HMI is interfaced with PLC? Write any two applications of HMI.

Q.25 What is relay? What are the advantages of PLC over relays? Explain the basic operations of PLC.

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**5th Sem/
Instrumentation & Control
Subject : PLC, DCS & SCADA**

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which of the following is not the part of PLC?
- a) Power supply b) CPU
 - c) Memory d) None of the above
- Q.2 _____ refers to a dashboard or screen used to control machinery.
- a) HMI b) SCADA
 - c) PLC d) DCS
- Q.3 Basic operation of PLC includes
- a) Scan input b) Execute program
 - c) Update output d) All of the above
- Q.4 The _____ instruction moves a copy of one data file word to a specific destination.
- a) ADD b) MOV
 - c) SUB d) DIV

- Q.5 DCS refers to
- a) Direct control system
 - b) Distributed communication system
 - c) Distributed control system
 - d) Digital control system
- Q.6 The process the PLC follows to read inputs, execute the control program and update outputs is known as
- a) Sinking b) Scan cycle
 - c) Sourcing d) None of the above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Expand SCADA.
- Q.8 Define HMI.
- Q.9 PLC does not have complex wiring compare to relay. (T/F)
- Q.10 For high level supervision of machine and process, SCADA is used. (T/F)
- Q.11 The _____ instruction in PLC adds the content of two sources.
- Q.12 Define scan cycle in PLC.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Write any four limitations of relays.
- Q.14 Explain the principle of working of PLC.
- Q.15 What is sinking and sourcing? Explain with example.
- Q.16 What are the timer instructions in PLC? Explain retentive timer.
- Q.17 Draw and explain the block diagram of DCS.
- Q.18 Write any four differences between HMI and SCADA.
- Q.19 What are the selection criteria of HMI?
- Q.20 Write down any four applications of SCADA.
- Q.21 What is comparison instruction? Explain equal and greater instruction.
- Q.22 Write any four differences between PLC and DCS.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Draw and explain the block diagram of PLC. Write any four applications of PLC.